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Leveraging Acoustic Features and Deep Neural Architectures for Audio Deepfake Detection

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***Abstract*—This research presents a comparative analysis of various audio features and high-level architectures for deefake detection with emphasis on computational efficiency. Several light-weight models are proposed as opposed to GAN-based approaches in literature for evaluating custom-generated deepfakes. The model is trained on Fake or Real dataset and achieved commendable performance using MFCC-Conformer and MFCC-LSTM feature-model combinations by achieving 87.61% and 87.52% accuracy, respectively. Specifically, the MFCC-Conformer recorded a TN of 526 and a FN of 18, along with an AUC score of 0.96, while the MFCC-DenseNet achieved a TN of 535, an FN of 9, and an AUC score of 0.96, underscoring their effectiveness in identifying fake audios. The outcomes underscore the effectiveness of the proposed models in combating the proliferation of misleading media content.**

***Index Terms*—Audio Deepfake detection, Audio features, MFCC, Conformer, LSTM, DenseNet**

I. Introduction

Audio deepfakes are synthetic audio created using sophisticated technologies to accurately replicate human voices. These can deceive listeners, making it hard to distinguish between real and fake audio. Based on a global McAfee survey conducted in 2023, it was found that 10% of individuals had been victims of scams involving AI voice cloning, and 77% of those who were targeted experienced financial losses [1]. Financial fraud, reputational damage, manipulating governments through misinformation and national security risks [2] are a few threats posed by deepfakes. Alongside these threats, audio deepfakes also have genuine applications, such as recreating the voices of deceased singers, enhancing advertising and helps provide synthetic voices for patients who have lost their ability to speak [3].

Substantial research has focused on audio applications, investigating acoustic features such as MFCC and STFT, and utilizing high-level architectures like CNN and RNN [4], [5]. GANs [6] have also been employed for their ability to discern subtle differences between genuine and synthetic audio. However, a significant challenge persists due to the computational demands of GAN-based models and the necessity to explore a diverse range of modern high-level architectures and audio features. Furthermore, there is a need to test the models against both custom-generated deepfakes and real-time audio. Addressing these challenges is critical to advancing the efficacy and accessibility of technologies for detecting audio deepfakes.

Prevailing models designed for audio deepfake detection are characterized by high computational demands and a limited scope of audio feature and architecture combinations tested against custom-generated deepfakes. In contrast, the proposed model addresses these challenges by prioritizing computational efficiency while achieving robust predictive accuracy with a modest dataset. Unlike conventional approaches reliant on GANs or simpler algorithms like SVM, this model explores advanced architectures such as Conformer and LSTM [7]. Furthermore, it utilises a diverse array of audio features including MFCC, mel spectrogram, ZCR, and chroma-based features [8]. Key contributions of this research include:

- Developing a bespoke model for computationally efficient audio deepfake detection trained on a subset of the FoR Dataset [9].
- Evaluation of the proposed model using various feature-model combinations by employing benchmark performance metrics.
- Comparing the proposed model with cutting-edge models in literature for audio deepfake detection

Section II offers a comprehensive review of existing literature, while Section III delves into the system architecture. Subsequently, Section IV showcases the findings and evaluates performance. Lastly, Section V summarises the study by outlining future prospects.

II. Related Work

Considerable research has been dedicated to detecting both video and audio deepfakes in various contexts. The focus of this review is on state-of-the-art audio deepfake detection models in literature. The study in [11] employs MFCC features extracted from both real and deepfake audio samples and using transfer learning by leveraging pre-trained models to enhance performance. Their approach demonstrates an impressive accuracy of 88.2% in identifying deepfake audio and a sensitivity of 85.8%, indicating its effectiveness in discerning synthesized from genuine audio content. The study in [12] adopts a

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feature-based approach where audio data is transformed into various spectral features such as MFCCs, chroma, spectral contrast, and tonnetz. Machine learning algorithms like SVM and random forest are used for classification. Additionally, an image-based approach is employed, converting audio samples into mel spectrograms processed by TCNs and STNs. This method achieves a test accuracy of approximately 92% in identifying deepfake audio.

In [13], the system utilizes the Conformer architecture, which integrates Transformers and CNNs to capture both local and global features in audio data. Experimental results on the ASVspoof 2021 Deepfake dataset demonstrate the system's performance, achieving an Equal Error Rate (EER) of 15.71%. The study described in [14] introduces a novel approach employing mel-spectrogram image features alongside Wavelet-CNN and VGG16 architectures to process spectral information critical for detecting spoofing. Their method achieves an EER of 2.8% on the ASVspoof 2019 challenge evaluation set. Research detailed in [15] employs the FoR dataset, consisting of four distinct subsets with varying audio lengths and processing characteristics. MFCC features are extracted and utilized to train a VGG-16 CNN model, achieving 93% test accuracy on the for-original subset.

Current methodologies face various limitations arising from a lack of extensive experimentation with combinations of high-level architectures and audio features. Moreover, these approaches have not been thoroughly tested on custom-generated deepfakes. This study aims to propose a computationally efficient model tailored for audio deepfake detection by exploring diverse combinations of high-level architectures and audio features, evaluating their efficacy against custom-generated deepfakes.

III. Methodology

The pipeline of the research is detailed in Fig. 1. The steps are described in the following subsections.

A. Dataset Description

The FoR-2sec dataset utilized for this research, is one of the four subsets of the FoR (Fake or Real) dataset [9]. The dataset also comprises custom generated deepfakes, generated by the authors using StyleGAN architecture [16]. The detailed description of the dataset is provided in the accompanying Table I. In this research, the FOR-2sec dataset was chosen over a more robust dataset like ASVspoof [10] to emphasize the primary objective of comparing the performance of various audio-based models while maintaining computational efficiency. By using a smaller dataset, we aimed to explore how different models perform in scenarios where computational resources are limited, making it more feasible for practical, real-world applications without compromising model evaluation across different architectures. This approach allowed us to balance accuracy with efficiency in our experimentation.

B. Data Preprocessing

The preprocessing of audio data involved several crucial steps to ensure the audio samples were prepared for training our model. First, the audio files were loaded and resampled to a consistent sample rate of 22,050 Hz using the librosa library [17]. Further, MFCCs, mel spectrograms, chroma, LPC and ZCR features, which capture the power spectrum of audio were extracted [18]. Given the varying durations of audio files, the MFCC arrays were padded with zeros to a maximum length, ensuring consistent input dimensions for the model. The labels for the audio files were transformed into one-hot encoded vectors. Finally, the input data was transposed to match the expected shape for the sequence-to-sequence model.

TABLE I
Dataset Description

Fake or Real Dataset (for-2seconds)	
Attribute	**Description**
Total no of audio samples	17,870
No of audio samples (Train)	13,956
No of audio samples (Test)	1,088
Custom Generated Audio Deepfake samples	10
No of audio samples (Validation)	2,826
Audio Length (per audio sample)	2 secs

C. Audio Based Features

Extracting audio features is essential for processing and analyzing audio signals, especially in audio classification and detecting deepfakes. The key audio features employed in our study are listed below [8], [11], [15], [18]:

1) **MFCC**: Captures the spectral characteristics of the audio signal. MFCC is generated by computing the power spectrum using FFT, passing it through a mel filterbank, and applying DCT to yield MFCC coefficients.
2) **Mel Spectrogram**: Represents the frequency representation of audio on the mel scale. It is created by computing the power spectrum with FFT, and converting it to the mel scale using a mel filterbank.
3) **ZCR**: Measures the rate at which audio signal crosses the zero amplitude axis within a specific time frame. A higher rate indicates more frequent signal transitions, typical of percussive or noisy sounds.
4) **Chroma**: Captures the harmonic content of audio by representing the twelve distinct pitch classes. They are robust to variations in timbre and dynamics.
5) **LPC**: Estimates the basic spectral characteristics by predicting the current sample value based on past samples. LPC coefficients are used in speech synthesis.

D. Training

In audio deepfake detection, training a robust model relies on extracting key features from audio signals and employing advanced architectures to discern intricate patterns. These features serve as crucial inputs for high-level architectures such as CNNs [19], Conformers, BiLSTMs, etc [4] which analyze the features to understand the intricate patterns in real and fake audio, thereby aiding in discrimination. Post-training, the model's weights are saved in *.h5 files. In this research, sparse categorical crossentropy was used as the

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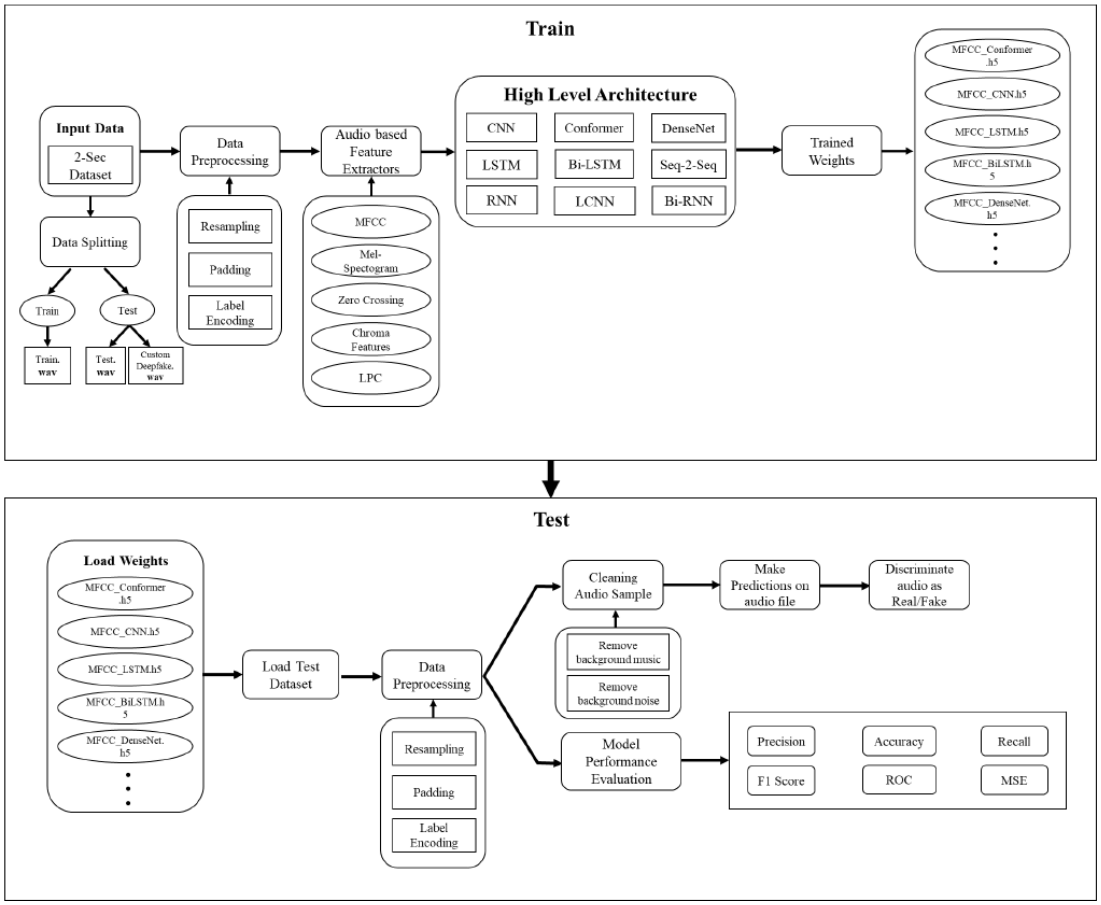

Fig. 1. Proposed architecture for Audio Deepfake Detection

loss function across all models. The high level architectures experimented with are highlighted below. Details regarding model architecture are available in [4], [7], [13], [19].

1) **LCNN**: Combines CNNs and LSTMs. They are effective in deepfake detection by analyzing visual and temporal aspects of multimedia content.
2) **RNN**: RNNs adeptly seize temporal dependencies in sequential data. In deepfake detection, they examine temporal patterns to identify anomalies.
3) **LSTM**: Captures long-term dependencies in sequential data, making them ideal for speech recognition and time-series forecasting. In deepfake detection, they analyze temporal patterns to identify manipulation.
4) **BiLSTM**: Process data in both forward and backward directions. This improves deepfake detection by identifying temporal inconsistencies.
5) **Conformer**: Integrates CNNs and self-attention to capture both short- and long-range relationships in sequential data. They analyse local and global patterns.
6) **Densenet**: Densely connected layers in Densenet promote effective feature reuse and smooth gradient flow. Densenets examine visual features to identify anomalies.
7) **GRU**: Simpler and more efficient than LSTMs. They are used in deepfake detection to analyze temporal patterns.

In this research, several audio classification models were employed with consistent hyperparameters across most models. The LCNN, RNN, LSTM, BiLSTM, Conformer, DenseNet, and GRU models all used the Adam optimizer with a learning rate of 0.001, a dropout rate of 0.5, ReLU activation for dense layers, and a Softmax activation for the output layer. Additionally, the models had a final dense layer of size 64, except for LCNN, which utilized a larger dense layer of 128 and also included a 0.25 dropout rate for the Conv2D layers. All models had a final dense layer with Softmax activation corresponding to the number of output classes.

E. Testing

For the testing phase, the test audio files were loaded and resampled to a uniform sample rate of 22,050 Hz using the librosa library [17].

IV. Results & Discussion

The proposed model entails experimenting with various combinations of high-level architectures and audio features to distinguish and identify audio deepfakes. Evaluation metrics including accuracy, precision, recall, F1-score, and the area under the Receiver Operating Characteristic curve (AUC). The models were tested against a dataset comprising 1088 audio

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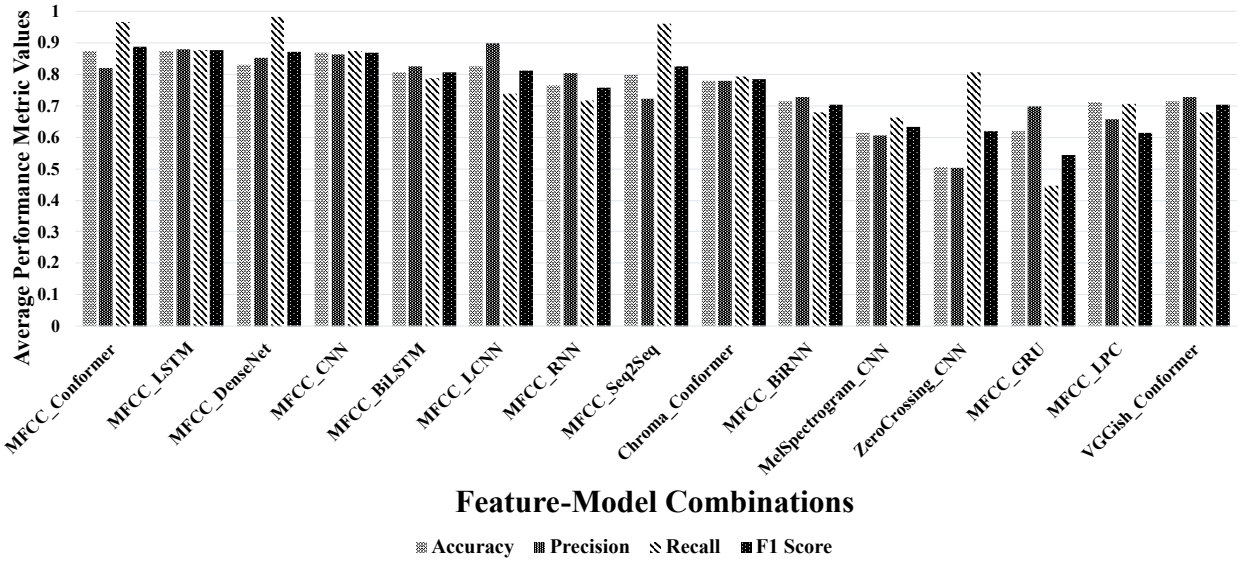
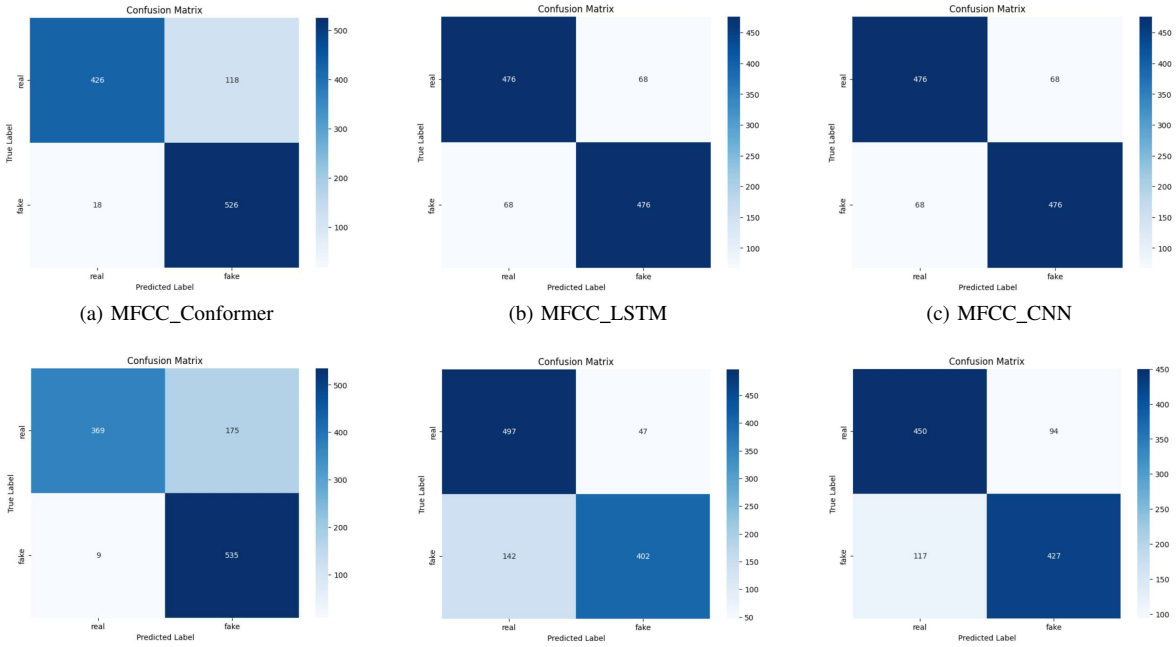


Fig. 2. Results of performance evaluation conducted on test audio data



(a) MFCC_Conformer (b) MFCC_LSTM (c) MFCC_CNN

(d) MFCC_DenseNet (e) MFCC_LCNN (f) MFCC_BiLSTM

Fig. 3. Confusion Matrix of Top 6 Feature-Model combinations viz. MFCC using Conformer, LSTM/BiLSTM,LCNN and Densenet. These combinations are identified based on the performance metrics plot as provided in Fig. 2.

TABLE II
BEST 6 MODELS BASED ON F1 SCORE

Model	Accuracy	Precision	Recall	F1 Score
MFCC_Conformer	0.88	0.82	0.97	0.89
MFCC_LSTM	0.88	0.88	0.88	0.88
MFCC_DenseNet	0.83	0.85	0.98	0.87
MFCC_CNN	0.87	0.86	0.88	0.87
MFCC_LCNN	0.82	0.89	0.74	0.81
MFCC_Seq2Seq	0.80	0.72	0.96	0.82

samples in the ”for-2 seconds” test set. The results observed for all the models tested on custom-generated deepfakes and dataset-based test set are highlighted in Fig.2. The performance of the best models in terms of F1 score is provided in Table II. From Table II, it can be observed that the trends are similar, except for the Seq2Seq model, with a higher F1-score than bi-LSTM model.

From Fig. 2, it can be observed that MFCC_Conformer combination achieved an accuracy of 87.61%, precision at 82.14%, recall at 96.61%, and an F1-score at 88.79%.

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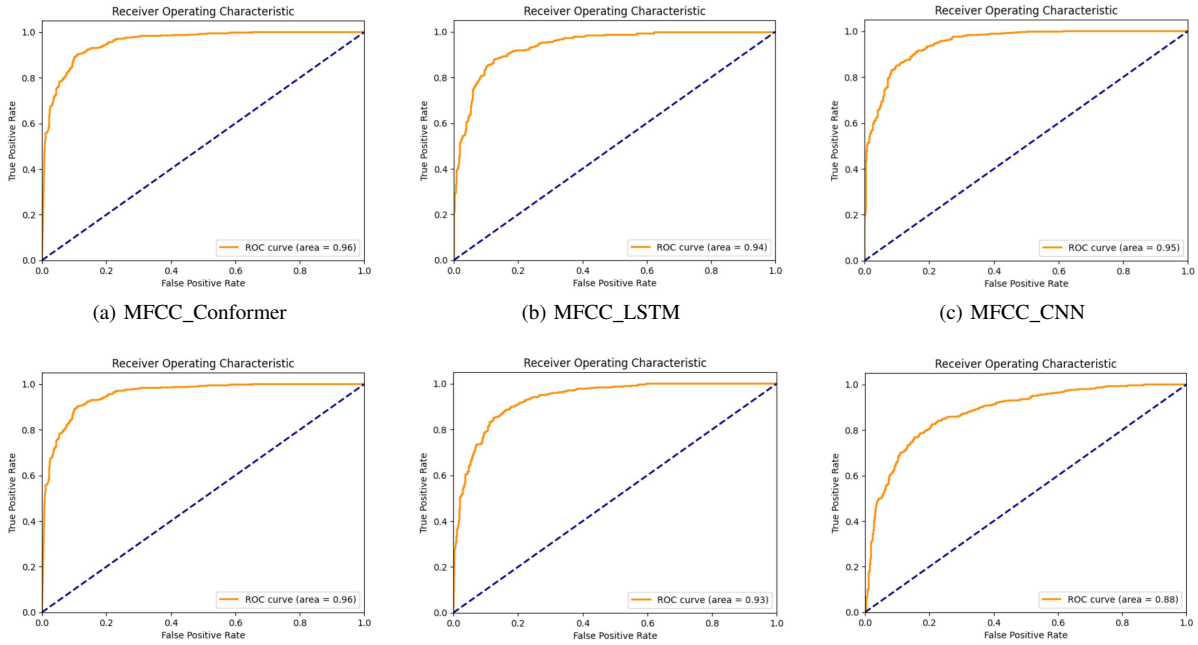

(a) MFCC_Conformer (b) MFCC_LSTM (c) MFCC_CNN

(d) MFCC_DenseNet (e) MFCC_LCNN (f) MFCC_BiLSTM

Fig. 4. ROC of Top 6 Feature-Model combinations viz. MFCC using Conformer, LSTM/BiLSTM,LCNN and Densenet

TABLE III
COMPARATIVE STUDY WITH STATE-OF-THE-ART

Year	Model	Tested On Unseen Deepfake?	No. of Features	Accuracy	Test Dataset
2019 [20]	CNN-VGG16	-	1	81.6%	FaceForensics++
2020 [9]	Timbre Model	Yes	4	NB- 67.27%, SVM- 73.46%, DT- 70.26%, RF- 71.47%	Fake or Real (FoR)
2020 [21]	STFT & Mel	-	3	STFT- 47.55%, Mel- 41.04%, MFCC- 88%	Fake or Real (FoR)
2021 [22]	MFCC & SVM	-	1	67%	Fake or Real (FoR)
2021 [23]	SVM & RF	-	1	75%	ASVspoof2019
2022 [24]	AdaBoost	-	1	85.28%	Fake or Real (FoR)
2023 [25]	CNN	Yes	2	86%	ASVspoof2019
2024	**Proposed Model**	**Yes**	**5**	**MFCC_Dense: 87.88%, MFCC_LSTM: 87.52%, MFCC_Conformer: 87.61%, MFCC_CNN: 86.85%**	**Fake or Real (FoR) & Custom Generated Deepfakes**

MFCC_LSTM model attained an accuracy of 87.52%, precision at 87.85%, recall at 87.54%, and an F1-score of 87.70%. MFCC_CNN model achieved an accuracy of 86.85%, precision at 86.38%, recall at 87.50%, and an F1-score at 86.94%. Meanwhile, MFCC_DenseNet model reached an accuracy of 83.08%, precision at 85.35%, recall at 98.34%, and an F1-score of 87.16%. MFCC_BiLSTM model obtained an accuracy of 80.65%, precision at 82.46%, recall at 78.64%, and an F1-score of 80.51%. Finally, MFCC_LCNN model demonstrated an accuracy of 82.45%, precision at 89.82%, recall at 73.84%, and an F1-score at 81.05%.

The confusion matrices and ROC curves of the top 6 models is highlighted in Fig.3 and Fig.4 respectively. For deepfake detection, it is crucial to identify fake content accurately, meaning true negatives (TN) should be high and false negatives (FN) should be low. The MFCC_Conformer model recorded a TN of 526 and an FN of 18, while the MFCC_DenseNet model had a TN of 535 and an FN of 9.

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The MFCC_Seq-2-Seq model recorded a TN of 523 and an FN of 21. Both the MFCC_LSTM and MFCC_CNN models had a TN of 476 and an FN of 68. The areas under the ROC curves (AUC) further substantiate the efficacy of our models: the MFCC_Conformer and MFCC_DenseNet models both had an AUC of 0.96, indicating excellent performance. The MFCC_LSTM and MFCC_CNN models had AUCs of 0.94 and 0.95, respectively. The MFCC_LCNN model had an AUC of 0.93, while the MFCC_BiLSTM model had an AUC of 0.88.

In summary, MFCC-based models, particularly those utilizing Conformer, DenseNet, LSTM, CNN and BiLSTM architectures, demonstrated superior performance across various evaluation metrics. The features extracted by the deep models are able to discriminate between real and fabricated audio in a computationally efficient manner.

The efficacy of the proposed model is underscored through a juxtaposition with contemporary methodologies, as depicted in Table III. This comparison specifically addresses the research gaps highlighted in this study, focusing on factors such as the variety of audio features employed, testing on custom-generated deepfakes, model performance and the datasets used. The models in literature are mainly trained on FoR or ASVspoof dataset. The maximum accuracy in literature is reported to be 88% using MFCC, but it is important to note that the model is not tested on custom deepfake, as opposed to our proposed model. This analysis emphasizes the critical role of selecting suitable acoustic features and architectures to elevate the precision and dependability of audio deepfake detection systems.

V. Conclusion & Future Scope

The study on audio deepfake detection systematically compared various feature and high-level model architectures. MFCC-based models outperformed the other features tested, owing to its alignment with human auditory perception. Among the high-level architectures explored, feature-model combinations such as MFCC-Conformer, MFCC-LSTM, MFCC-DenseNet, MFCC-CNN, MFCC-BiLSTM, and MFCC-LCNN showed exceptional performance in distinguishing real from fake audio samples. These findings underscore the importance of selecting appropriate audio feature and high level architectures for robust and accurate detection.

Future research should explore additional audio features like Constant-Q Transform and Wavelet Transform alongside MFCCs. Investigating advanced model architectures such as transformers, capsule networks, and attention mechanisms can enhance detection accuracy. Larger, more diverse datasets are crucial for training models to generalize across different deepfake audio types, ensuring more reliable detection systems.

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